

# Object Detection Extension

<https://github.com/aiyshawary/Object-Detection-Extension-in-Rapid-Miner>

## OVERVIEW

A RapidMiner Studio extension that brings transfer learning-based object detection into a no-code environment for training and inference with Faster R-CNN, FCOS, SSD, SSDLite, and RetinaNet models.

## IMPACT

Developed two RapidMiner operators for object detection — `FineTuneObjectDetectionModel` and `ObjectDetectionModelInference` — to enable users to train and deploy detection models within a visual workflow without writing code.

## KEY DETAILS

1. `FineTuneObjectDetectionModel` accepts image and annotation directories (YOLO/Pascal VOC) and CSV.
2. `ObjectDetectionModelInference` runs predictions on new images and returns bounding boxes and class probabilities.
3. Supports five architectures: Faster R-CNN, FCOS, SSD, SSDLite, and RetinaNet for varying speed-accuracy trade-offs.
4. Configurable for CPU-only and GPU (CUDA cu118) environments; installed as a JAR extension in RapidMiner.
5. Enables domain-specific object detection with minimal ML expertise using a drag-and-drop process.
6. Inference operator can optionally save annotated images with overlaid prediction bounding boxes via a `DataFrame` output.

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7. Uses chardet for automatic annotation-file encoding detection, ensuring robust parsing of YOLO/Pascal systems and locales.